Reflection

1. The initial import “java.awt.event.ActionListener” is not used, so I remove it.

2. Adding “outputField.setText(“”); to refresh every unit boxes whenever the user clicks on a new measurement (line 68)

```
66      // Dynamic dropdown updates based on category
67      private void updateUnitBoxes() {
68          outputField.setText("");
69          sourceUnitBox.removeAllItems();
70          targetUnitBox.removeAllItems();
```

3. The previous JComboBox did not declare the specific type that the box held, so it will print warnings like uncheck/unsafe operations. After adding String, the parameter tells the compiler that the box holds String (line 7-12, 30, 41-45)

```
7   private JComboBox<String> categoryBox;
8   private JTextField inputField;
9   private JComboBox<String> sourceUnitBox;
10  private JTextField outputField;
11  private JComboBox<String> targetUnitBox;
12  private JButton convertButton;
```

```
30          categoryBox = new JComboBox<>(categories);
```

```
41          inputField = new JTextField("1");
42          sourceUnitBox = new JComboBox<>();
43          outputField = new JTextField();
44          outputField.setEditable(false);
45          targetUnitBox = new JComboBox<>();
```

4. When the user tried to input something invalid such as “abc”, the code previously put 請輸入有效數字 in the output field. This might confuse user as the answer, adding error message and visual cue help distinguish this error (line 121-124)

```
121         } catch (NumberFormatException ex) {
122             outputField.setText("");
123             JOptionPane.showMessageDialog(this, "請輸入有效數字", "輸入錯誤", JOptionPane.ERROR_MESSAGE);
124         }
```